Styrene Butadiene Styrene modified asphalt waterproofing sheet

Group number: M50

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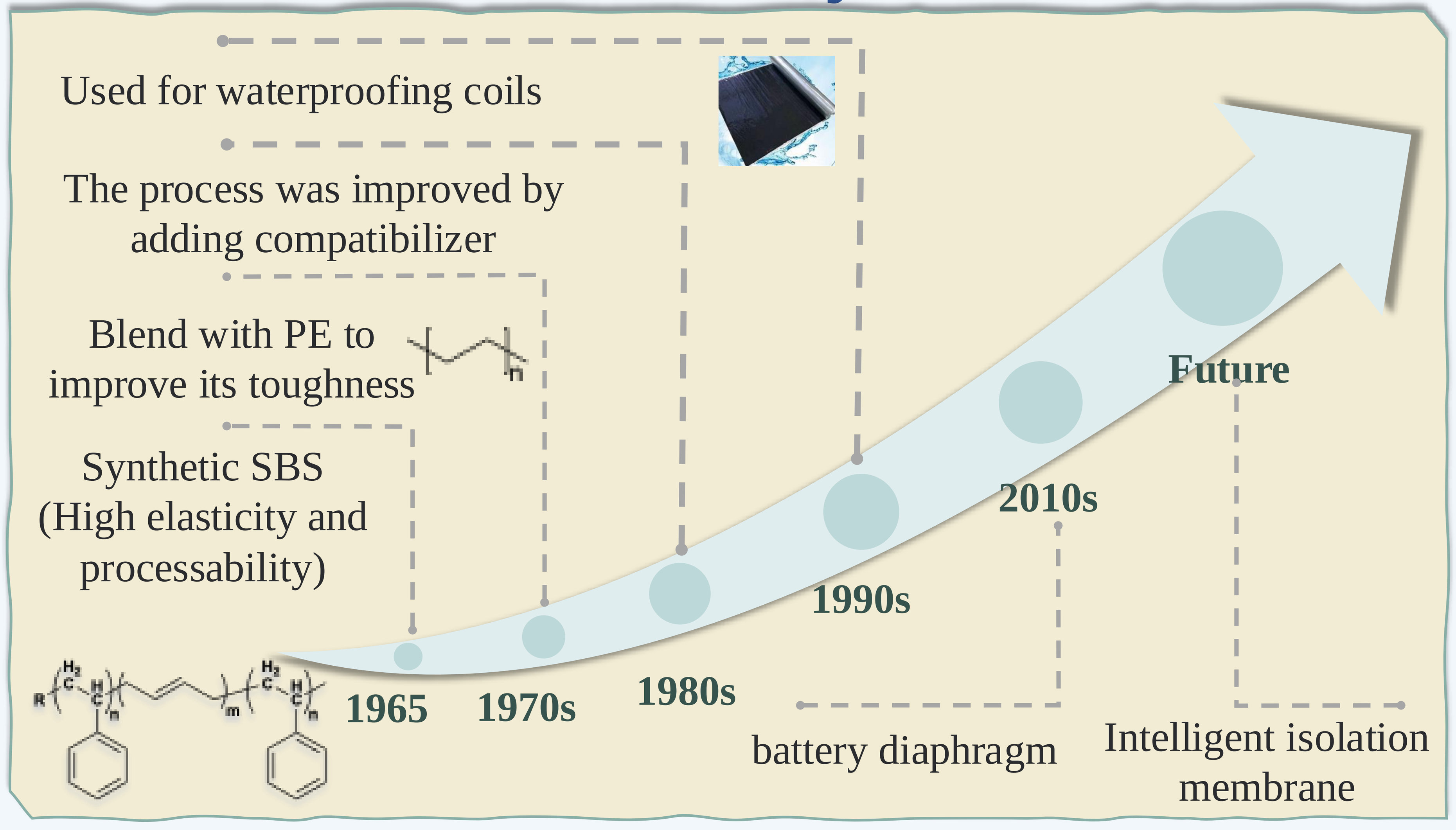
1 Abstract

SBS (Styrene-butadiene-styrene block copolymer), as a thermoplastic elastomer with excellent properties, is mainly synthesized by anionic polymerization and coupling method. Its molecular structure is unique, with both the high elasticity of rubber and the plasticity of plastic, showing rubber characteristics at room temperature and can be melted like plastic at high temperatures**. Especially in the field of roads, SBS can significantly improve asphalt performance, and improve pavement resistance to rutting**, fatigue and weather resistance. With its outstanding comprehensive performance, SBS plays a key role in numerous industrial and everyday application scenarios.

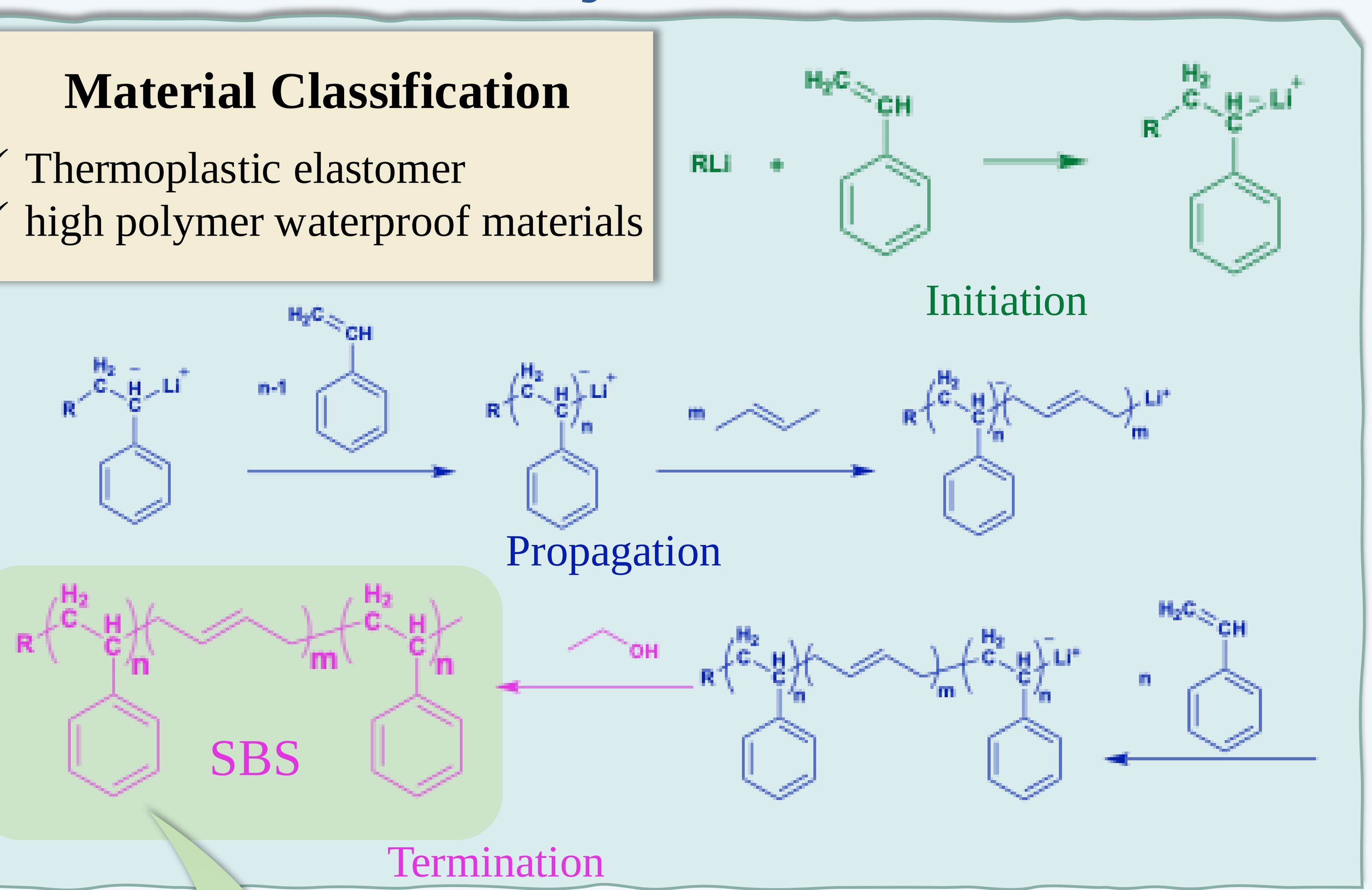
2 History



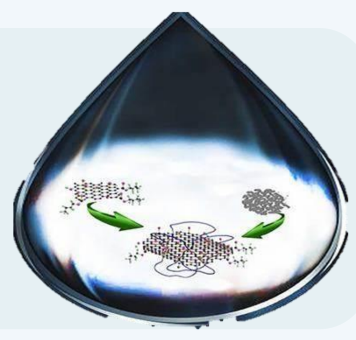
3 Synthesis

Material Classification

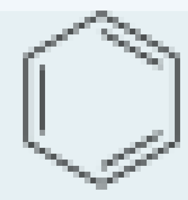
- ✓ Thermoplastic elastomer
- ✓ high polymer waterproof materials



After blending, the asphalt forms a uniform and continuous film layer, reducing interface defects and reducing water permeability.



4 Features: Waterproof



Benzene ring is hydrophobic

5 Application--Asphalt was improved with SBS

Raise softening point: Enhances the ability of asphalt to resist deformation at high temperature, reduces the possibility of rutting, hugging and other diseases in the high temperature of the road surface

Reduces high-temperature viscosity: Make asphalt easier to spread and compact during construction

✓ Improved high temperature performance

Increase ductility: Withstand greater deformation at low temperatures without cracks

Reduced brittleness point: Extend the service life of the pavement

✓ Improved low temperature performance

Application site [2]

- ✓ Road engineering
- ✓ Bridge engineering
- ✓ Airport engineering

Figure 1. The process of SBS modified asphalt.

6 Property Profiles

Table 1. The property profiles of SBS [2]

Property	Index
Density	0.9-0.95 g/cm³
Shore hardness	60-90A a certain softness
Elasticity	The elastic recovery rate is above 90%
Breathability	Gases and water vapor can pass through
Electrical	10^{12} - $10^{15}\Omega \cdot$ cm

7 Safety Information

Toxicity:

- It is not classified as dangerous goods and poses no direct health hazard to humans.
- Data from long-term toxicity studies are lacking.

Potential Risks:

- Harmful combustion gases or vapors may be released in a fire.
- Severe reactions that may occur with strong

Storage and transportation:

- Tightly closed and dry.

[1]Haozongyang Li, Chengwei Xing, Bohan Zhu, Xiao Zhang, Yang Gao, Shixian Tang, Huailei Cheng, Comparative analysis of four styrene-butadiene-styrene(SBS) structure repair agents in the rejuvenation of aged SBS-modified bitumen,Construction and Building Materials,Volume 476,2025,141232,

[2] Song Xu, Xiangjie Niu, Shaoxu Cai, Bingtao Xu, Lei Fang, Investigation on active rejuvenation mechanism of aged SBS modified bitumen: Insights from experiments and molecular dynamics,Polymer Degradation and Stability, Volume 238, 2025, 111336, ISSN 0141-3910, https://doi.org/10.1016/j.polymdegradstab.2025.111336.